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LEARN TRADE GROW

A STOCK MARKET LEARNING & TRADING SIMULATION PLATFORM

A full-stack web application designed to teach users the fundamentals of investing through structured lessons and quizzes that assess their understanding. The platform provides a safe environment where users can practise trading using a real-time simulator powered by live market data.

Karine Gassanova

BSc (Hons) in Software Systems Development

Department of Computing and Mathematics

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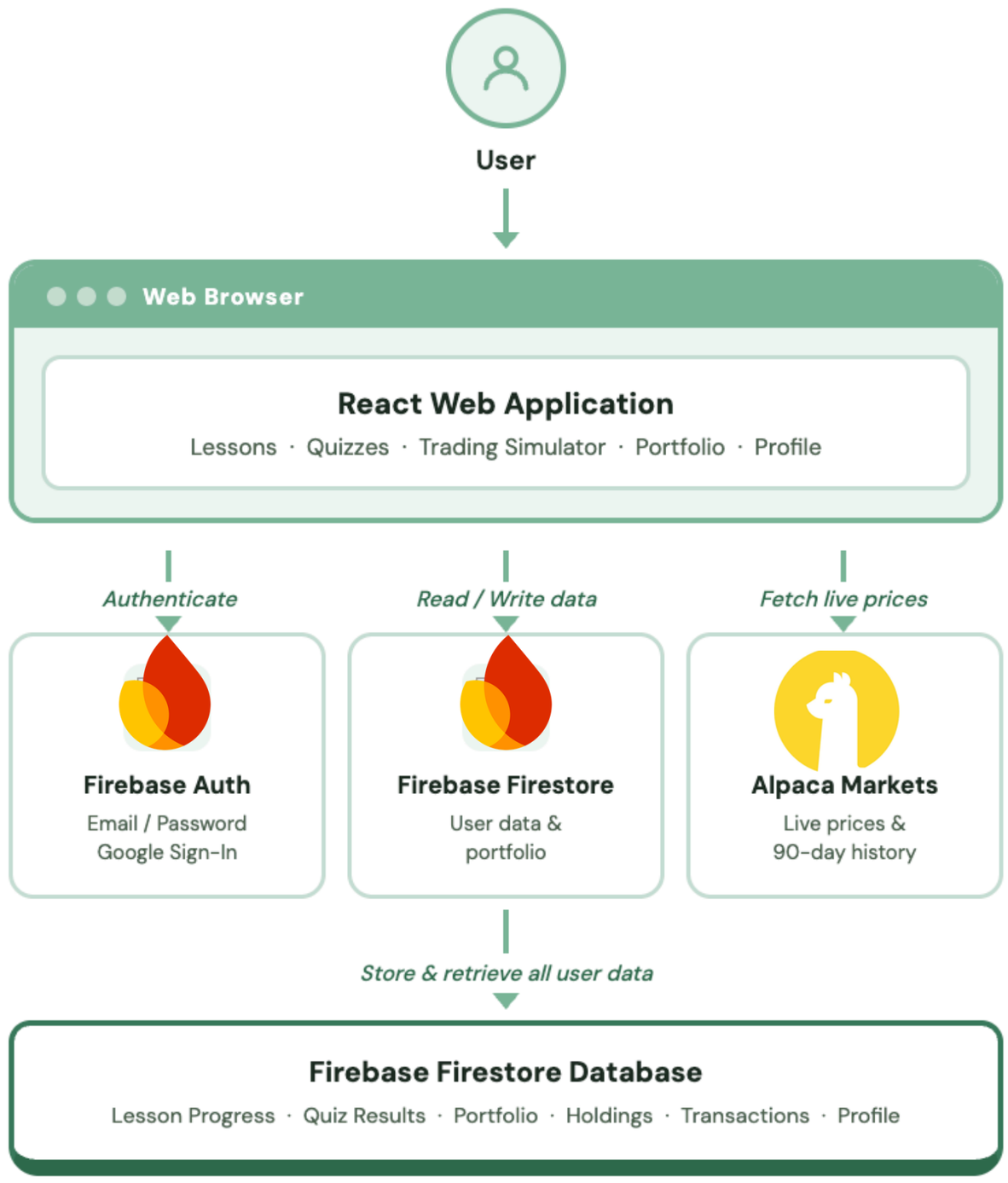
OVERVIEW

An interactive web platform designed to help users develop practical investment knowledge through eight structured lessons covering stock market fundamentals, chart analysis, risk management and trading strategies. Each lesson concludes with a quiz requiring 70% to progress. Users then apply their knowledge in a live trading simulator using €10,000 of virtual funds.

TECHNOLOGY STACK

React — Front-end Framework	JS — Core Language	Firebase — Auth & Database
Development IDE	GitHub — Version Control	Alpaca — Live Stock Market Data API

SYSTEM ARCHITECTURE



METHODOLOGY

Developed using an Agile approach across four time-boxed sprints, each involving planning, development, testing and review.

Plan Goals › **Build** Develop › **Review** Test › **Repeat** Next

1. **Authentication & Database** — Jan–Feb
2. **Lessons & Quiz Engine** — Feb–Mar
3. **Trading Simulator** — Mar
4. **Testing & Finalisation** — Mar–Apr

KEY FEATURES

Structured Learning Path
Eight lessons from stock market basics to advanced strategies, unlocked step by step.

Gated Quiz System
A 70% pass mark is required after each lesson before the next stage is unlocked.

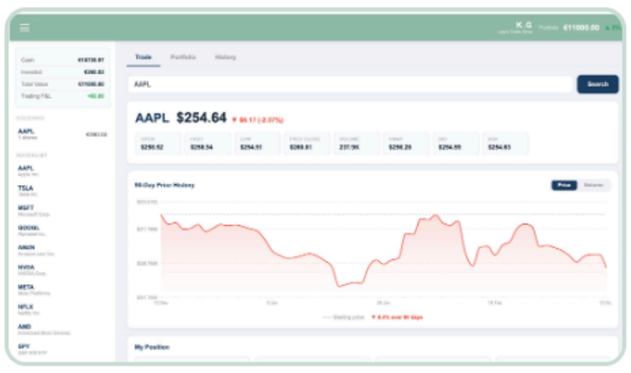
Trading Simulator
Trade real stocks using €10,000 of virtual funds with live Alpaca market data.

Live Price Charts
90-day interactive chart showing price trends and daily volume bars.

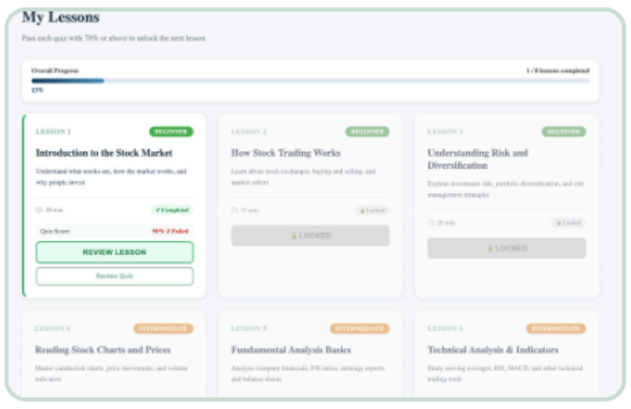
Portfolio Tracker
View holdings, total invested, portfolio value and P&L updated in real time.

User Profiles
Custom photo, avatar colour and display name synced across every page.

APPLICATION INTERFACE



Trading Simulator



Lessons Page